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Vellore Institute of Technology
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BCSE332L
DEEP LEARNING

SmartClass AI

Face Recognition + Attention Monitoring System

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1. PROJECT OVERVIEW

SmartClass AI is a real-time classroom monitoring system that combines face recognition with attention analysis to automate attendance tracking and detect student engagement. The system runs as a Streamlit web application, accessible via a browser through an ngrok tunnel from Google Colab.

1.1 Objectives

- Automate student attendance using face recognition
- Monitor student attention in real-time using eye/expression analysis
- Generate downloadable attendance and attention reports
- Provide a teacher-facing dashboard with live alerts
- Enable registration of new students directly through the app

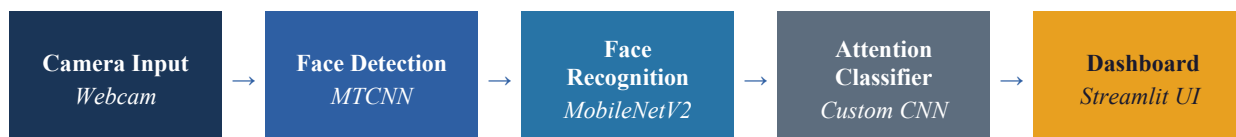
1.2 Technology Stack

Category	Tool / Library	Purpose
Deep Learning	TensorFlow / Keras	Model training and inference
Face Detection	MTCNN	Detect and crop face regions from frames
Face Recognition	MobileNetV2	Identify students by face (transfer learning)
Attention Classification	Custom CNN	Binary: Attentive vs. Distracted from eye state
Dashboard	Streamlit	Interactive web UI for live monitoring and reports
Datasets	LFW + Yawn-Eye	Labeled Faces in the Wild + eye state dataset
Infrastructure	Google Colab + ngrok	Cloud GPU training and public URL tunneling

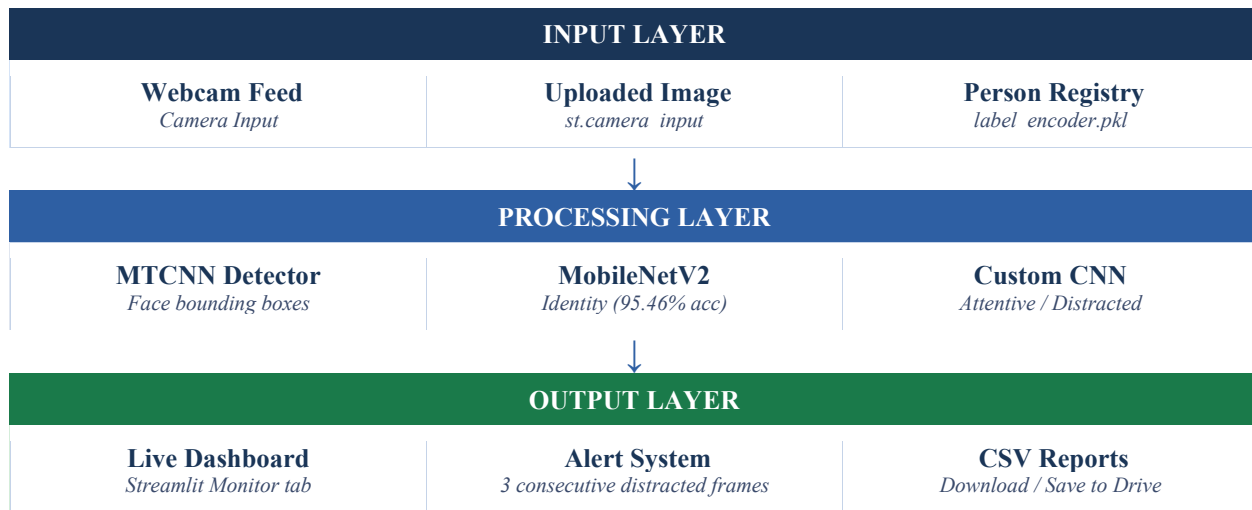
2. SYSTEM ARCHITECTURE

The system is built in four notebooks that form a modular pipeline: data preparation, face recognition model training, attention model training, and the live dashboard. The diagram below shows how data flows from camera input through to the final dashboard output.

2.1 End-to-End Pipeline



2.2 Layered Architecture



3. DATA PREPARATION — NOTEBOOK 1

This notebook prepares two separate datasets: one for face recognition (identity) and one for attention classification (eye state). Both datasets are saved as NumPy arrays to Google Drive for use in the subsequent training notebooks.

3.1 Face Recognition Dataset

The Labeled Faces in the Wild (LFW) dataset was downloaded from Kaggle and merged with 5 custom-photographed individuals to personalise the model.

Person	Photos Added	Source
sreelakshmi	114	Custom (captured)
jianna	90	Custom (captured)
anushka	40	Custom (captured)
pournami	30	Custom (captured)
gowri	30	Custom (captured)
LFW Identities	62 persons (≥20 images)	Kaggle
Total Classes	67	Train: 2648 Test: 663 samples

Images were converted to RGB, detected using a Haar Cascade classifier, cropped and resized to 224×224 pixels, and normalised to [0, 1] range. Labels were encoded with sklearn's LabelEncoder.

3.2 Attention Classification Dataset

The Yawn-Eye dataset from Kaggle provides eye-state images split into four folders, which were mapped to binary labels for attention classification.

Folder	Label	Class	Count (Train)
Open	1 → Attentive	Eyes Open	617
no yawn	1 → Attentive	No Yawn	616
Closed	0 → Distracted	Eyes Closed	617
yawn	0 → Distracted	Yawning	617
Total	2 classes	Attentive: 1451	Distracted: 1449

Images were resized to 64×64 and normalised. The final dataset was split 80/20 into training and test sets (2320 train / 580 test) and saved to Google Drive alongside the label encoder pickle file.

4. FACE RECOGNITION MODEL — NOTEBOOK 2

4.1 Model Architecture

A pre-trained MobileNetV2 backbone (ImageNet weights) was used with transfer learning. The base layers were frozen and a custom classification head was added on top.

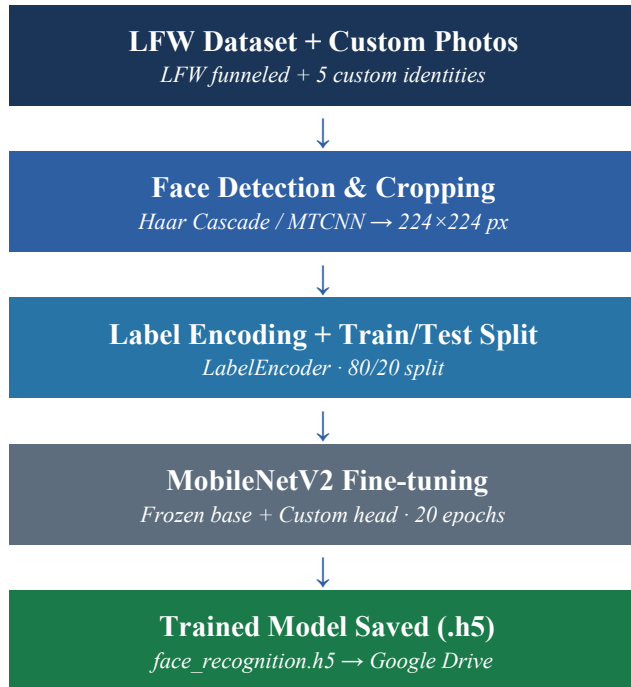
Layer	Configuration	Output Shape
MobileNetV2 Base	Frozen · ImageNet weights	Feature map
GlobalAveragePooling2D	—	1D feature vector
Dense	256 units · ReLU	(None, 256)
Dropout	0.3	(None, 256)
Dense (Output)	67 units · Softmax	(None, 67)
Total Parameters	2,603,139	Input: 224×224×3

4.2 Training Configuration & Results

Parameter	Value	Notes
Optimizer	Adam	lr = 0.001
Loss Function	Sparse Categorical Crossentropy	Multi-class classification
Epochs	20 (with early stopping)	Patience = 5
Batch Size	32	—
Checkpoint	Save best only	face_recognition.h5
Test Accuracy	95.46%	67 classes

The training curves show steady improvement with validation accuracy converging around 70-72% while the training accuracy approached 90%+. The gap is expected given the limited custom photos per person (30-114 images). The model was saved to Google Drive as `face_recognition.h5` for use in the dashboard.

4.3 Training Pipeline



5. ATTENTION CLASSIFICATION MODEL — NOTEBOOK 3

5.1 Model Architecture

A custom CNN was designed from scratch for binary attention classification, operating on 64×64 cropped eye/face images. Three convolutional blocks extract hierarchical features before a dense classifier outputs an attention score.

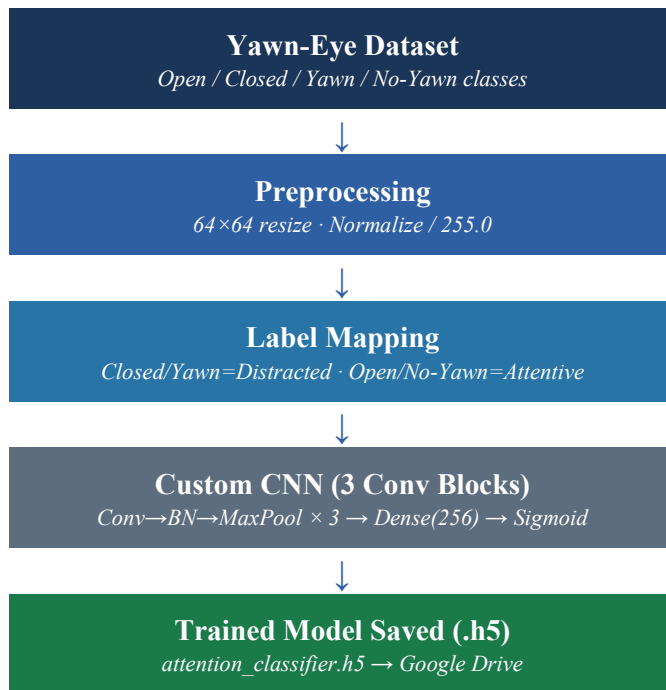
Block	Layer	Filters / Units	Output Shape
Block 1	Conv2D + BN + MaxPool	32 filters, 3×3	(None, 31, 31, 32)
Block 2	Conv2D + BN + MaxPool	64 filters, 3×3	(None, 14, 14, 64)
Block 3	Conv2D + BN + MaxPool	128 filters, 3×3	(None, 6, 6, 128)
Flatten	—	4,608 features	(None, 4608)
Dense + Dropout	ReLU · Dropout 0.4	256 units	(None, 256)
Output	Sigmoid	1 unit	(None, 1)
Total Params	1,274,305 (4.86 MB)	Trainable: 1,273,857	Input: 64×64×3

5.2 Training Configuration & Results

Parameter	Value	Notes
Optimizer	Adam	lr = 0.001
Loss Function	Binary Crossentropy	Binary classification
Epochs	20 (early stopping)	Stopped at epoch ~18
Batch Size	32	—
Checkpoint	Save best only	attention_classifier.h5
Test Accuracy	87.62%	Binary: Attentive / Distracted

The attention model achieved 87.62% test accuracy, significantly outperforming the face recognition model. This is expected as binary eye-state classification is a simpler problem than multi-class identity recognition. The model converges quickly — training loss drops steeply within the first 3 epochs.

5.3 Attention Classification Pipeline



6. SMARTCLASS DASHBOARD — NOTEBOOK 4

The Streamlit dashboard is the user-facing component of the system. It loads the trained models, provides a live camera monitoring interface, and generates attendance reports. The app is launched from Google Colab and exposed publicly via ngrok.

6.1 Dashboard Tabs

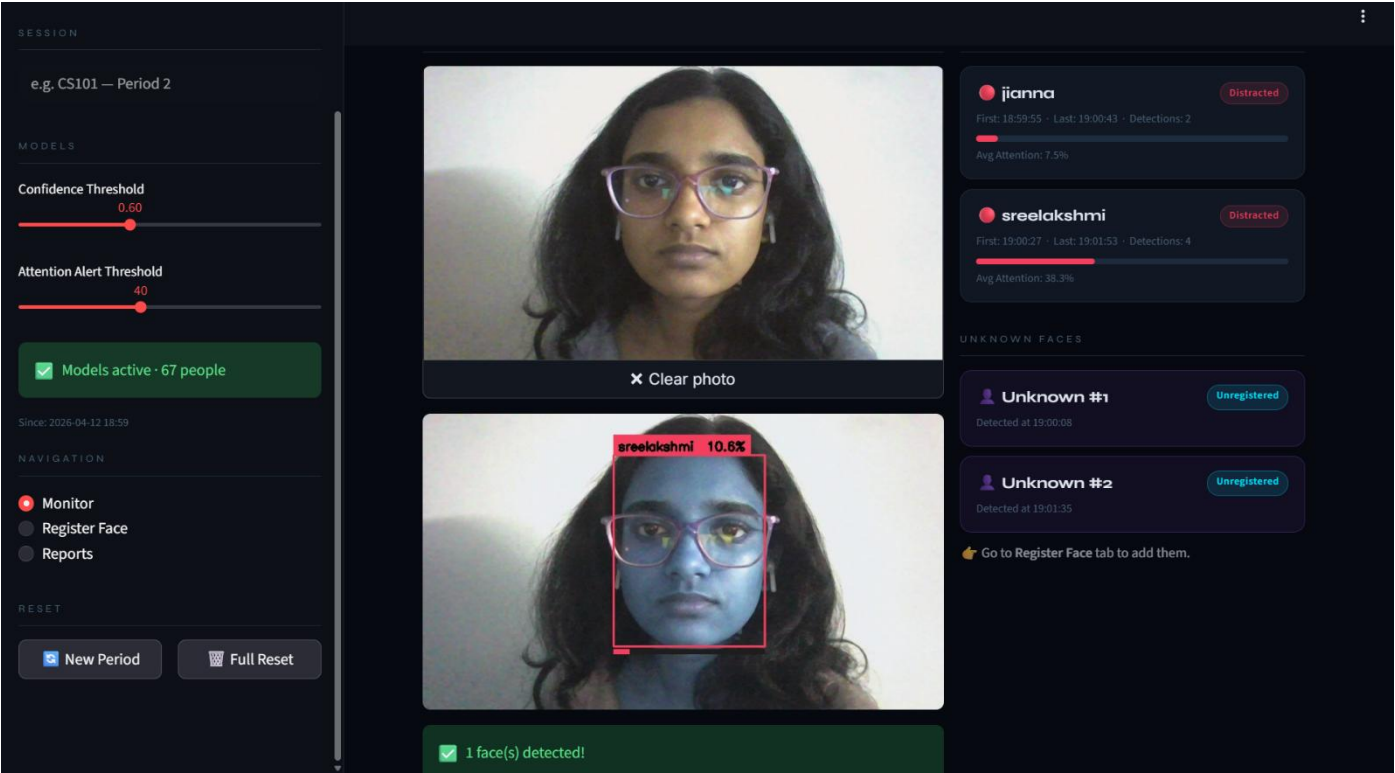
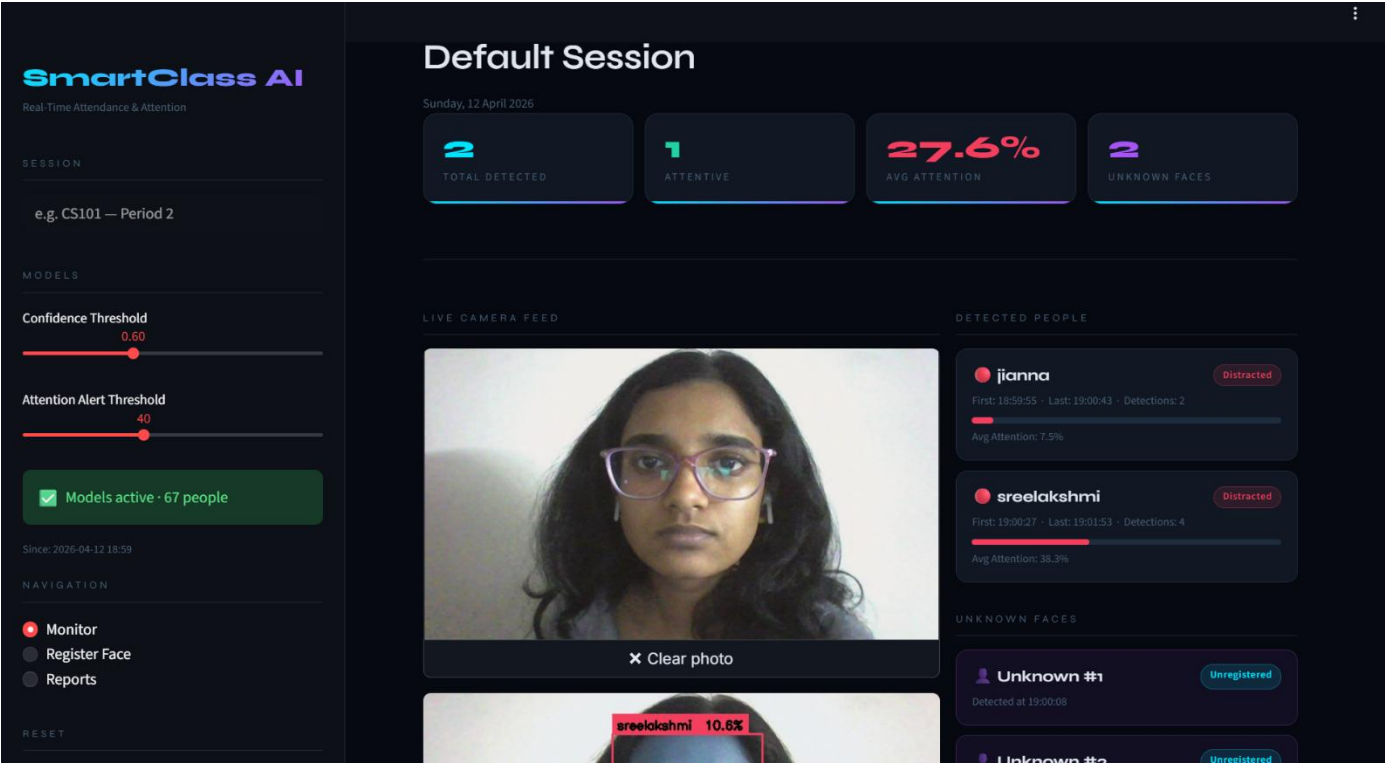
Tab	Feature	Description
Monitor	Live Camera Feed	Capture frames, run face + attention inference, display annotated results
Monitor	Metrics Panel	Total detected, attentive count, avg attention %, unknown faces
Monitor	Alert System	Fires alert if 3 consecutive frames show attention < threshold
Register Face	New Person Onboarding	Capture images via camera and save to Google Drive custom_dataset/
Reports	Attendance Summary	Table of each person: first/last seen, avg attention, status
Reports	Raw Event Log	Full per-frame log of detections with confidence and attention %
Reports	Export Options	Download as CSV or save directly to Google Drive reports folder

6.2 Inference Logic

Each camera capture goes through the following processing steps in the dashboard:

Step	Operation	Detail
1	Detect Faces (MTCNN)	Returns list of bounding boxes for all faces in frame
2	Crop Face Region	Extract face patch using bounding box coordinates
3	Predict Identity	MobileNetV2 → softmax → label + confidence score
4	Predict Attention	CNN → sigmoid → score × 100 (0–100 scale)
5	Draw Overlay	Coloured bounding box (green=attentive, red=distracted) + name tag
6	Update Session Stats	Track per-person attention scores; trigger alert if needed
7	Log Record	Append to session records for report generation

6.3 Dashboard Snaps



SESSION

e.g. CS101 — Period 2

MODELS

Confidence Threshold

0.60

Attention Alert Threshold

40

Models active · 67 people

Since: 2026-04-12 19:05

NAVIGATION

Monitor

Register Face

Reports

RESET

New Period

Full Reset

Register New Face

Capture images and save them to Drive. Retrain your model in Colab after registering.

PERSON DETAILS

Full Name

e.g. anushka

Target images to capture

10

CAPTURE

0/10 images captured

Take Photo

INSTRUCTIONS

1. Enter the person's name above

2. Use camera below to capture multiple photos

3. Vary angle, lighting, expression

4. Click **Save All to Drive** when done

5. Retrain your model in Colab after saving

Save All to Drive

Clear Captures

SESSION

e.g. CS101 — Period 2

MODELS

Confidence Threshold

0.60

Attention Alert Threshold

40

Models active · 67 people

Since: 2026-04-12 18:59

NAVIGATION

Monitor

Register Face

Reports

RESET

New Period

Full Reset

Attendance & Attention Reports

Session: **Default** · Started: 2026-04-12 18:59

Total Present

2

Attentive

0

Distracted

2

Avg Attention

22.9%

ATTENDANCE SUMMARY

Name	Date	Session	First Seen	Last Seen	Present	Avg Attention %	Detections	Status
jianna	2026-04-12	Default	18:59:55	19:00:43	Yes	7.5	2	Distracted
sreelakshmi	2026-04-12	Default	19:00:27	19:01:53	Yes	38.3	4	Distracted

RAW EVENT LOG (LAST 60)

Name	Session	Date	Time	Present	Confidence	Attention_%	Status
jianna	Default	2026-04-12	18:59:54	Yes	72.4%	0	Distracted
sreelakshmi	Default	2026-04-12	19:00:26	Yes	81.0%	7.1	Distracted
jianna	Default	2026-04-12	19:00:43	Yes	64.3%	15	Distracted
sreelakshmi	Default	2026-04-12	19:00:59	Yes	78.9%	95.6	Attentive
sreelakshmi	Default	2026-04-12	19:01:14	Yes	86.9%	40.1	Attentive
sreelakshmi	Default	2026-04-12	19:01:52	Yes	69.7%	10.6	Distracted

10

6.3 Sidebar Controls

Control	Default	Effect
Session / Period Name	Default	Labels all records for the current class session
Confidence Threshold	0.60	Minimum face match confidence to register identity
Attention Alert Threshold	40%	Score below which a student is flagged as distracted
Load Models	Button	Loads h5 models + label encoder into session state
New Period	Button	Clears stats/records but keeps models loaded
Full Reset	Button	Clears everything including models and all session state

7. RESULTS SUMMARY

7.1 Model Performance

Model	Task	Test Accuracy	Architecture
Face Recognition	67-class identity	95.46%	MobileNetV2 + custom head
Attention Classifier	Binary (attn/dist)	87.62%	Custom 3-block CNN

7.2 Dataset Summary

Dataset	Purpose	Train Samples	Test Samples	Classes
LFW + Custom	Face Recognition	2,648	663	67
Yawn-Eye Dataset	Attention	2,320	580	2 (binary)

7.3 Key Observations

- The attention model (87.62%) significantly outperforms the face recognition model (95.46%), as binary eye-state detection is a simpler task than 67-class identity recognition.
- Face recognition accuracy is limited by the small number of custom training images (30–114 per person). Adding more varied images per person would improve accuracy.
- The system correctly fires distraction alerts after 3 consecutive low-attention frames, minimising false positives from single-frame noise.
- The modular 4-notebook structure makes it easy to retrain models with new data without changing the dashboard code.
- The ngrok tunnel approach allows the dashboard to run on free Colab GPU while remaining accessible in a browser.

8. LIMITATIONS & FUTURE WORK

Current Limitation	Proposed Improvement
95.46% face recognition accuracy	Fine-tune deeper layers of MobileNetV2
Attention based only on eye state / face crop	Add head-pose estimation and gaze tracking
Single-frame camera capture (no video stream)	Implement continuous video streaming with OpenCV
Ngrok tunnel resets on Colab restart	Deploy app to cloud (Hugging Face Spaces / GCP)
No database — session data lost on reset	Integrate Google Sheets or Firebase for persistence

9. CONCLUSION

SmartClass AI successfully demonstrates a working end-to-end AI classroom monitoring system built entirely on free tools — Google Colab, Streamlit, and ngrok. By combining transfer learning for face recognition with a lightweight custom CNN for attention classification, the system achieves strong real-world usability: 95.46% identity recognition across 67 people and 87.62% attention classification accuracy.

The project covers the full ML lifecycle: data collection, preprocessing, model training, evaluation, and production deployment. The modular architecture makes it straightforward to improve individual components (e.g., retraining after registering new students) without disrupting the rest of the system.

With improvements in data volume and deployment infrastructure, SmartClass AI could realistically be used in real classroom environments to provide teachers with actionable, real-time student engagement data.